



Q4

Create a risk matrix (asset, threat, vulnerability, likelihood, impact, risk)

5 Marks

Objective: Build a simple risk matrix that rates each asset-threat pair by likelihood and impact and derives an overall risk level.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	List the key assets involved in the breach.	Asset inventory	Asset list
2	ANALYSE	For each asset, name the threat and the vulnerability it exploited.	Threat/vuln notes	Threat + vuln per asset
3	TEST	Rate likelihood as Low / Medium / High.	Likelihood scale	Likelihood set
4	TEST	Rate impact as Low / Medium / High.	Impact scale	Impact set
5	ANALYSE	Combine likelihood × impact to get the risk level and colour it.	$Risk = L \times I$	Colour-coded risk
6	REPORT	Sort the matrix so the highest risks appear first.	Sort by risk	Prioritised matrix

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Risk matrix (Acme HealthCare)

Asset	Threat	Vulnerability	Likelihood	Impact	Risk
Patient DB	Data theft	Weak email filter	High	High	CRITICAL
Booking portal	Ransomware	Unpatched server	Medium	High	HIGH
Staff accounts	Credential theft	No MFA	High	Medium	HIGH

Reading the matrix: the Patient DB is the top priority (High likelihood and High impact = Critical), so its mitigations are addressed first in Q5.

Reusable Template

The [2023 MGM Resorts cyberattack](#) stands out as one of the most high-profile social engineering and ransomware events in recent years. The incident completely paralyzed the operational systems of prominent Las Vegas casino resorts, proving that human-centric defenses are just as critical as technological firewalls.

- **Attack Date:** The attack formally began on September 8, 2023, and systems were aggressively brought down by ransomware deployment on September 11, 2023.
- **Threat Actor:** Carried out by the Scattered Spider group (acting as an access broker affiliate) in coordination with the ALPHV/BlackCat ransomware organization.
- **Initial Access Route:** Attackers used LinkedIn to scrape organizational structure data, located an employee, and called the MGM corporate IT help desk. Through a 10-minute voice phone call (vishing), they successfully tricked the help desk staff into bypassing security parameters and resetting the target's Multi-Factor Authentication (MFA) credentials.

Risk matrix template (fill in; the Risk cell auto-colours by level)



Asset	Threat	Vulnerability	Likelihood	Impact	Risk
Identity Access Control (Okta/Active Directory)	Privilege escalation & administrative hijacking	IT Help Desk Vishing (Resetting user MFA based on basic corporate profile info)	High	High	CRITICAL
Resort Infrastructure (Hypervisors / ESXi Servers)	Comprehensive ransomware encryption	Over-privileged accounts allowing full management control across cloud endpoints	High	High	CRITICAL
Customer Operational Portals	Service blackout and severe	Flat internal connectivity	Medium	High	HIGH

Your Output / Screenshot (paste result here):

- **Operational Chaos:** For roughly 10 days, all interconnected systems across major Strip properties went dark. Digital hotel room keys failed, ATMs shut down, casino slot machines displayed blank error screens, and front desk staff had to handle check-ins completely by hand.
- **Financial Impact:** MGM Resorts sustained approximately \$100 million in lost revenue alongside an additional \$10 million in direct incident response and remediation costs